

HEALTHCONNECT CLINIC

Week 7 Cross-Track Testing & Validation | Data Analytics → Data Science

AnalystLab Africa Experience Lab Internship Programme

Project	HealthConnect Clinic Experience Lab
Week	Week 7
Primary Track	Data Analytics
Collaborating Track	Data Science
Integration Theme	Lead Time × Previous No-Show History
Status	Analytics-side validation completed; further Data Science response pending

Purpose

This document records the Week 7 continuation of the Data Analytics–Data Science collaboration established in Week 6. It uses the existing Week 6 collaboration as the starting point and documents the Analytics testing completed in Week 7. It does not claim a Data Science model improvement that has not been confirmed.

Week 6 Collaboration Carried Forward

During Week 6, Data Analytics identified a combined high-risk segment: appointments booked more than 30 days in advance with previous no-show history recorded a 67.9% observed no-show rate, compared with 55.2% among appointments booked more than 30 days in advance without previous no-show history. The overall no-show baseline was 48.5%.

Data Science indicated that booking lead days and historical no-show rate were treated as independent signals in its modelling approach. The Analytics finding therefore led to a candidate Lead Time × Previous No-Show interaction feature for testing.

Week 7 Collaborative Testing

Component / Dependency	Analytics Testing	Result	Status
30+ days + prior no-show	Revalidated using dashboard and slicer testing	67.9% observed no-show rate	Validated
30+ days + no prior no-show	Revalidated using dashboard and slicer testing	55.2% observed no-show rate	Validated
Lead Time × Previous No-Show relationship	Compared both segments with 48.5% overall baseline	Combined segment remained analytically distinct	Descriptively validated

Candidate interaction feature	Confirmed analytical inputs remain supported	Predictive effect not established	Pending model test
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Test → Finding → Action → Retest

Stage	Evidence	Action	Outcome
Test	Week 6 interaction finding carried into Week 7	Recheck segments using refined dashboard filters	Rates remained consistent
Finding	67.9% vs 55.2% within the 30+ day group	Retain interaction as modelling dependency	Candidate remains analytically supported
Action	Analytics-side validation completed	Document finding for cross-track use	Traceable handoff maintained
Retest / external validation	No further Data Science response available	Do not infer model improvement	Open dependency documented

Information Provided to Data Science

- Overall no-show baseline: 48.5%.
- 30+ Day No-Show Rate: 60.5%.
- 30+ days with previous no-show history: 67.9%.
- 30+ days without previous no-show history: 55.2%.
- Interpretation as observed association rather than causal effect.
- A candidate Lead Time × Previous No-Show interaction hypothesis for predictive testing.

Information Received from Data Science

The Week 6 response from Data Science was that booking lead days and historical no-show rate were treated as independent signals and that an explicit interaction feature could be tested. The team indicated that the interaction would be considered as a candidate feature for Week 7 testing.

No additional model-performance result was available for this submission. This is recorded as a pending dependency rather than replaced with an assumed outcome.

What Changed for the Project

The collaboration established a clear analytical-to-predictive handoff: two descriptive signals were converted into a specific interaction hypothesis that Data Science can evaluate through its modelling process. Week 7 strengthened the evidence by revalidating the underlying segment in the refined dashboard.

Evidence and Communication Record

The original collaboration was documented in the Week 6 Cross-Track Collaboration Report. The Week 7 Analytics-side validation is supported by the Week 7 Testing & Validation Evidence Report. The collaboration communication record can also be referenced through the Slack conversation used for the handoff:

Collaboration Limitation

Because no further response was available from the Data Science collaborator at the time of submission, this document does not claim that the interaction feature was implemented, improved model performance, or produced a better prediction. The evidence supports the Analytics finding, the candidate interaction hypothesis, and the Analytics-side revalidation.

Next Integration Step

The remaining cross-track step is for Data Science to evaluate the Lead Time × Previous No-Show interaction using its established modelling and evaluation process. Any statement about predictive usefulness should be based on the resulting model-comparison evidence.